Anthony James Padavano

SOFTWARE ENGINEER, AGENTS & MULTI-AGENT SYSTEMS

New York City | 561-602-7300 | padavano.anthony@gmail.com

LINKS

Portfolio · GitHub · agentic-titan · ORGANVM System

PROFILE

Designed and operates agentic-titan, a polymorphic multi-agent orchestration framework with 9 switchable topologies (pipeline, hierarchy, swarm, ring, star, mesh, tree, consensus, hybrid), 845 tests, and 18 completed development phases — built to solve the coordination problems existing frameworks consistently undersolve: fixed topology, model lock-in, and absent production safety infrastructure. Separately, agent--claude-smith implements multi-agent orchestration using the Claude Agent SDK with subagent spawning, permission-scoped tool whitelists, atomic session persistence, and governance-first spawn validation. The AI-conductor methodology — human directs architecture, AI generates volume, human reviews — enabled one person to build 113 repositories across 8 GitHub organizations with 23,470 verified tests, 104 CI/CD pipelines, and zero dependency violations.

TECHNICAL SKILLS

Multi-Agent Orchestration · Agentic Systems Design · LLM API Integration · Claude Agent SDK · MCP Protocol · Eval Harness Development · Prompt Engineering · Context Window Management · Model Selection · Python · TypeScript · FastAPI · Agent Registry Architecture · Session State Management · Tool Design · Subagent Spawning · Governance-First Architecture · CI/CD · GitHub Actions · Observability & Instrumentation

SELECTED PROJECTS

agentic-titan — Polymorphic Multi-Agent Orchestration Framework
Model-agnostic, self-organizing multi-agent system with 9 switchable coordination topologies (pipeline, hierarchy, swarm, ring, star, mesh, tree, consensus, hybrid) that switch at runtime based on task state — not fixed at design time. 845 tests across adversarial, chaos, e2e, integration, performance, MCP, and Ray test suites. 18 completed development phases. Human-in-the-loop approval gates, role-based access control, budget enforcement, and audit logging built as structural requirements. Python. ORGAN-IV (Taxis).

agent--claude-smith — Claude Agent SDK Orchestration with Subagent Spawning
Multi-agent orchestration using the Claude Agent SDK: a central Orchestrator spawns purpose-built subagents (code reviewer, task executor, security auditor, AI bridge), each with its own session, tool permissions, retry budget, and security constraints. Enforces acyclic spawn graphs, bounded concurrency, per-agent tool whitelists, and command validation before any shell operation executes. Atomic file-backed session persistence. Chezmoi-based TOML templates for operator-defined agent definitions. TypeScript. ORGAN-IV (Taxis).

organvm-mcp-server — 88-Tool System Graph Exposure via MCP Protocol
MCP server (stdio transport) exposing the full 113-repository system graph to AI agent sessions: 88 tools across 16 groups (registry, seeds, graph, health, governance, sessions, SOPs, metrics, coordination). Enables agents to query dependency graphs, check promotion state, inspect seed contracts, and coordinate work programmatically. The infrastructure layer that lets AI agents navigate the system without filesystem access.

ORGANVM Eight-Organ System — 113-Repository AI-Conductor Infrastructure
8-organization orchestration system coordinating 113 repositories through registry-v2.json (single source of truth), seed.yaml contracts (produces/consumes edges), automated dependency validation (50 edges, 0 violations), and a formal promotion state machine. 23,470 verified tests across 20 flagship repos. 104 CI/CD pipelines. ~739K words of documentation. Built by one person using the AI-conductor methodology: human directs architecture, AI generates volume, human reviews every output. Python + TypeScript.

Application Pipeline — Agentic CLI Tooling Suite
Python CLI system with 59 standalone commands and MCP server integration for autonomous pipeline execution. Scoring rubric, campaign orchestrator, YAML state machine (research → qualified → drafting → staged → submitted), batch enrichment, Greenhouse/Lever/Ashby ATS submitters, and an agent mode that previews and executes pipeline actions autonomously. ~1,977 tests. Shared pipeline_lib.py foundation decomposed into focused modules.

EXPERIENCE

Independent Engineer — Agentic Systems & Multi-Agent Orchestration 2020 — Present

- **Multi-Agent Frameworks:** Designed agentic-titan (polymorphic topology engine, 9 coordination patterns, runtime topology switching, 845 tests) and agent--claude-smith (Claude Agent SDK integration, subagent spawning, governance-first session management, acyclic spawn graph enforcement).
- **Eval & Observability:** Built adversarial, chaos, e2e, integration, performance, and MCP test suites for agent systems. FastAPI + HTMX observability dashboard with live health, dependency graph visualization, and system state instrumentation.
- **MCP Protocol:** 88-tool MCP server exposing system graph to AI agents — registry queries, seed inspection, dependency tracing, coordination primitives. The AI-conductor interface between human architecture decisions and AI execution volume.
- **Infrastructure at Scale:** 113 repositories, 104 CI/CD pipelines, 23,470 verified tests, 50 validated dependency edges, 0 violations — all designed and operated by one practitioner using structured AI-conductor methodology over 33 named development sprints.

Instructor — Composition & Writing 2015 — Present

Miami Dade College, Keiser University, Nova Southeastern, FAU, Young Harris, Tarrant County, Broward, Palm Beach State

- 100+ courses across 8 institutions; 2,000+ students. Documentation-first communication discipline applied to agent system design and technical writing.

Digital Marketing Manager 2023 — 2024

Miami Dade College Foundation (nonprofit)

- Rebuilt foundation website; designed campaigns increasing donor engagement by 32%.

Multimedia Specialist 2011 — 2020

AJP Media Arts — clients incl. Boca Ballet Theatre, Forward Funding, Arts in Boca

- Video, audio, and visual campaigns for nonprofits and arts organizations; 17.5M+ views, $2M in fundraising.

EDUCATION

M.F.A. Creative Writing, Florida Atlantic University 2015 — 2018

B.A. English Literature, CUNY College of Staten Island 2010 — 2014

CERTIFICATIONS

Full-Stack Developer, Meta (2024) · UX Design, Google (2023) · Digital Marketing, Google (2023) · Project Management, Google (2023)